

Bike Sharing Demand Forecasting - Exploratory Data Analysis Report

This report presents an Exploratory Data Analysis (EDA) and forecasting overview for the Bike Sharing Dataset from the UCI Machine Learning Repository. The objective is to predict hourly bike rental demand ('cnt') using weather, seasonal, and temporal features to support operational planning and bicycle logistics.

1. Business Problem

A local bicycle rental company needs reliable hourly demand forecasts to decide how many bicycles should be ready for rental at different times of the day and across the year. The operational goal is to reduce stock-outs during high-demand periods while avoiding unnecessary repositioning or over-preparation during low-demand periods. The forecasting target is cnt, the total hourly number of rentals.

2. Dataset Overview

The analysis uses the UCI Bike Sharing Dataset, which contains hourly rental records from Capital Bikeshare in Washington D.C. The dataset combines rental counts with calendar, season, and weather information. The project workflow loads the official hourly file, validates required columns, parses timestamps, sorts observations chronologically, and stores raw data, processed data, model artifacts, plots, notebooks, tests, and application code in separate folders.

- Target variable: cnt, total hourly rentals.
- Important calendar inputs: hour, weekday, working day, holiday, month, year, and season.
- Important weather inputs: weather situation, temperature, feeling temperature, humidity, and windspeed.
- Leakage variables excluded from modeling: casual and registered, because together they make up cnt.

3. Exploratory Data Analysis

The exploratory analysis focuses on operational demand patterns that matter for planning: daily trends, peak hours, seasonality, weather sensitivity, and relationships between weather variables and demand.

- Demand is strongly time dependent. Working days show commute peaks, while weekends and holidays have a broader daytime demand curve.
- Season matters. Fall and summer show higher average demand than spring.
- Weather matters. Clear weather has the highest average demand, while rain or snow reduces rentals.
- Demand is right-skewed: many hours have modest rentals, while peak periods require careful planning buffers.

4. Feature Engineering

Feature engineering is designed to support both model accuracy and production maintainability. The workflow keeps the feature list explicit, separates target leakage columns, and packages preprocessing with each trained model.

- Time features: hour and month are used directly, and cyclical sine/cosine features are added so the model can learn repeating time patterns.
- Encoding features: categorical variables such as season, hour, weekday, working day, and weather situation are one-hot encoded inside model pipelines.
- Scaled features: numeric variables are scaled for Linear Regression and passed through for tree-based models.
- Production benefit: the same pipeline handles preprocessing and prediction, reducing the risk of training-serving mismatch.

5. Forecasting Models

The supporting workflow compares multiple regression models: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. The final report presents one recommended model for the business case: XGBoost.

- Linear Regression is simple and explainable, but underfits nonlinear demand behavior.
- Random Forest captures nonlinear interactions and performs well, but the serialized model is larger.

- Gradient Boosting captures nonlinear effects with compact sequential trees.
- XGBoost provides the strongest accuracy in this workflow while remaining practical for a daily batch prediction service.

6. Model Performance

Model performance is evaluated on a chronological holdout period, which is more realistic for forecasting than a random split. The main business metric is mean absolute deviation, implemented as mean absolute error. It tells planners the average hourly bike-count miss.

Selected model	Mean absolute deviation	MAPE	R-squared
XGBoost	50.91	0.491	0.886

XGBoost is selected because it has the lowest mean absolute deviation: 50.91 hourly rentals. This means the model is off by about 50.91 rentals per hour on average in the holdout period.

7. Business Insights

- Peak demand is concentrated around commuting hours on working days, which is important for fleet readiness and station balancing.
- Weekend and holiday demand is more spread across the daytime, requiring a different operational profile than weekday commuting demand.
- Clear weather and warmer seasons support higher demand; poor weather reduces expected utilization.
- Hourly planning should use buffers around peak periods because the demand distribution includes high-volume spikes.

8. Operational Recommendations

The forecast should be used as a planning baseline for bicycle availability, staffing, and repositioning. The business should combine model output with local operational knowledge for events, disruptions, and planned service changes.

- Refresh forecasts daily after the latest completed rental data is available.
- Refresh intraday when weather forecasts change materially or actual demand deviates from plan.
- Use the model mainly for short- to medium-term planning where calendar and weather assumptions are credible.
- For longer-range annual planning, combine the model with fleet growth, station changes, pricing plans, holidays, and local event calendars.

9. Production Considerations

For a daily prediction service, the code should be maintainable by colleagues and safe to run repeatedly. The project therefore separates ingestion, feature engineering, training, evaluation, plotting, report generation, tests, and Streamlit UI code.

- Data quality: validate schema, missing values, target ranges, and timestamp parsing.
- Model governance: version model artifacts and record model comparison metrics.
- Monitoring: track forecast error by hour, weekday, season, and weather condition.
- Maintainability: keep reusable logic in source modules instead of duplicating code across notebooks.
- Testing: run unit tests before operational releases.

10. Conclusion

XGBoost is the recommended model for Part 1 because it achieves the best accuracy in the project workflow, with a mean absolute deviation of 50.91 hourly rentals. The model is suitable for a business planning case because it captures nonlinear demand behavior, can be refreshed in a production-style workflow, and gives planners an interpretable error metric for operational decisions.